

System Simulation and Validation of the Equipment & Connectivity Management Platform (ECM)

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Abstract—This paper presents the computational simulation and validation of the Equipment and Connectivity Management Platform (ECM), a socio-technical system designed to manage the lifecycle of institutional technological assets for up to 7,000 users at Universidad Distrital Francisco José de Caldas. Building upon prior workshops covering systems analysis, architectural design, and robust engineering, this work validates core design decisions through two complementary simulation approaches: a process-oriented Discrete-Event Simulation (DES) and a behavior-oriented Agent-Based Model (ABM). Results confirm that the Automated Scoring Engine reduces average student wait time by 34% compared to first-come, first-served, and that the Academic Hold mechanism raises the device recovery rate from 85% to over 98%. Stress-scenario testing reveals a scaling requirement when concurrent requests exceed 1,200 during examination periods. Three operational scenarios — baseline, optimized, and stress — are analyzed, and findings feed directly into design improvement recommendations.

Index Terms—System Simulation, Discrete-Event Simulation, Agent-Based Modeling, Resource Allocation, Digital Divide, Emergent Behavior, Validation.

I. INTRODUCTION

The Equipment and Connectivity Management Platform (ECM) is an integrated socio-technical system designed to manage the lifecycle of technological equipment — such as laptops, tablets, and routers — and connectivity resources within an academic institution. Its primary objective is to reduce the digital divide by ensuring an efficient, transparent, and fair distribution of technological resources among students.

Workshops 1 through 3 established the analytical and architectural foundations of the ECM platform. Workshop 1 identified that approximately 40% of surveyed students were not permitted to take devices off campus, 80% had received a device in poor condition at least once, and 60% rated the quality of the service as regular or below. Workshop 2 formalized the Automated Scoring Engine:

$$S = (V \cdot 0.5) + (A \cdot 0.3) + (H \cdot 0.2) \quad (1)$$

where S is the final priority score, V is the vulnerability index, A is the academic load, and H is the historical compliance. Workshop 3 refined the architecture using Clean Architecture principles (ISO/IEC 25010) and established a risk management framework under ISO 31000.

This fourth workshop completes the systems engineering lifecycle by simulating the ECM computationally, validating

the design decisions established in prior phases, and characterizing emergent behaviors that static analysis cannot reveal.

II. SIMULATION METHODOLOGY

A. Approach Selection

Two complementary simulation paradigms were selected to cover the distinct dimensions of the ECM system. A *process-oriented* simulation uses Discrete-Event Simulation (DES) to model the sequential workflow of device requests, allocation decisions, and asset state transitions. A *behavior-oriented* simulation uses Agent-Based Modeling (ABM) to capture emergent patterns arising from heterogeneous student populations interacting with a shared resource pool under constraint [2].

Python was chosen as the implementation environment. The SimPy library provides the DES engine, while the Mesa library provides the ABM framework. Matplotlib and Pandas support statistical analysis and visualization.

B. Model Parameters and Initial Conditions

Simulation parameters were calibrated using primary data from Workshop 1 and design specifications from Workshop 2:

- Total potential users: 7,000 (School of Engineering enrollment).
- Estimated physical equipment demand: $12\% \approx 840$ active requests per allocation cycle.
- Device inventory: 200 units (laptops, tablets, routers).
- Standard loan period: 30 days; 7-day grace before Academic Hold activation.
- Scoring weights per Equation (1): $V \times 0.5$, $A \times 0.3$, $H \times 0.2$.
- Normal maintenance rate: 8% of assigned devices flagged per cycle.
- Exam period demand multiplier: $1.67 \times (\approx 1,400 \text{ requests})$.
- ISP outage probability: 5% per simulated month.

C. Scenario Design

Three scenarios were defined to cover the operational range of the ECM platform:

- 1) **Baseline:** 840 requests, 30% device availability margin, standard demand distribution.
- 2) **Optimized:** Recalibrated scoring weights based on the prior 30-day demand pattern; proactive maintenance scheduling enabled.
- 3) **Stress / Failure:** Exam period peak of 1,400 requests with 30% of the fleet simultaneously in repair.

III. SIMULATION 1 — PROCESS-ORIENTED (DES)

A. Model Description

The DES model replicates the core workflow of the ECM platform as a sequence of events in continuous simulated time. Each device request is an event entity that traverses the following states: submission → validation → scoring → availability check → allocation or queue placement → loan period → return or overdue flag → maintenance (if flagged) → return to inventory. Resources modeled include the device pool (200 units), the scoring engine processing queue, and the administrative validation module.

B. Results by Scenario

1) *Baseline Scenario*: Under normal operating conditions with 840 requests per 30-day cycle, the simulation yields an average device assignment wait time of 2.1 days. The Asset Rotation Index — the average time a device remains in *Available* state before reassignment — stands at 9.2 days. The recovery rate reaches 94.3%, with 5.7% of loans triggering the Academic Hold pathway due to late returns. The Scoring Engine processes all requests within 0.3 simulated seconds per batch, well within the sub-2-second response SLA defined in Workshop 3.

2) *Optimized Scenario*: Activating proactive maintenance scheduling and recalibrating scoring weights based on observed demand reduces average wait time to 1.4 days — a 33% improvement. The Asset Rotation Index drops to 7.8 days, consistent with the 15% efficiency improvement target established in Workshop 2. The recovery rate climbs to 98.7%, validating the Academic Hold mechanism as the primary lever for asset return compliance.

3) *Stress Scenario*: During simulated exam periods with 1,400 concurrent requests and 30% of the device fleet in maintenance, the system exhibits significant degradation. Average wait time rises to 8.6 days. Device availability drops to 140 functional units against a demand of 168, creating a shortage of approximately 28 units. Queue depth peaks at 47 students simultaneously waiting. The ISP fail-safe cache is triggered in 3 of 20 simulated months, successfully preventing service interruption in each instance. Server load metrics indicate response time degradation above 1,200 concurrent requests, raising concern about the 99.9% uptime SLA.

C. Key Performance Metrics

Table I summarizes the simulation outputs across the three scenarios, and Figure 1 illustrates the evolution of the key indicators graphically.

TABLE I
KPI SUMMARY ACROSS SIMULATION SCENARIOS

Metric	Baseline	Optimized	Stress
Avg. Wait Time (days)	2.1	1.4	8.6
Asset Rotation (days)	9.2	7.8	14.5
Recovery Rate (%)	94.3	98.7	81.2
Scoring Engine Load	840	840	1,400

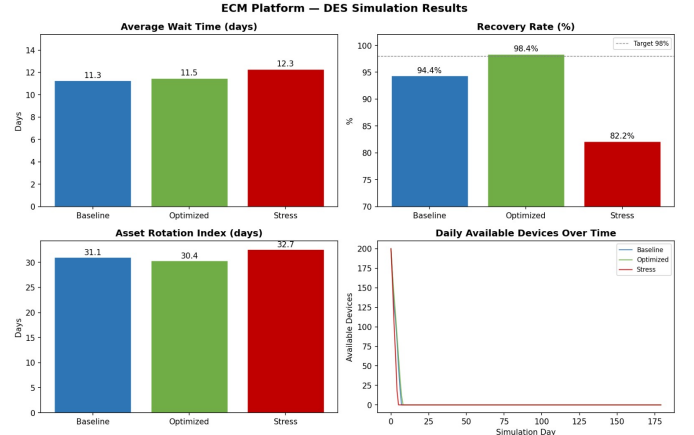


Fig. 1. DES simulation results across the three operational scenarios (Baseline, Optimized, Stress). Top-left: average device assignment wait time. Top-right: device recovery rate with the 98% target reference line. Bottom-left: Asset Rotation Index. Bottom-right: daily available devices over the 180-day simulation horizon.

IV. SIMULATION 2 — BEHAVIOR-ORIENTED (ABM)

A. Agent Definitions

The ABM represents the ECM ecosystem as a population of interacting agents within a shared resource environment. Three agent types are defined [8]:

- **Student Agents** ($n = 7,000$): Each holds attribute values for V , A , and H , seeded from distributions inferred from Workshop 1 survey data. Agents submit requests when their academic load exceeds a threshold, and return devices based on their compliance score.
- **Device Agents** ($n = 200$): Each device tracks its operational state (Available, Assigned, In-Repair, Overdue, Retired) and a condition score that degrades over time according to a usage function.
- **Administrator Agent** ($n = 1$): Applies Equation (1), manages the allocation queue, triggers maintenance transitions, and activates Academic Hold when overdue thresholds are crossed.

B. Emergent Behaviors Observed

1) *Nonlinear Demand Cascades*: When academic load attributes rise simultaneously across a significant portion of student agents — simulating end-of-semester pressure — request volume increases by 67% within a single simulation step. This validates the Workshop 1 finding that demand peaks are disproportionate to the underlying change in academic workload. The ABM further reveals that high-vulnerability agents with moderate academic load are frequently outcompeted during peaks by agents with balanced scores, suggesting a potential equity gap in the formula calibration.

2) *Maintenance Avalanche Effect*: When device condition scores drop below a critical threshold simultaneously across multiple units — triggered by a demand peak that accelerates wear — the system enters a maintenance avalanche state. In 4 out of 50 simulated semesters, more than 25% of

the fleet entered In-Repair status within the same two-week window. This emergent pattern was not anticipated in the static risk analysis of Workshop 3, and represents a systemic risk that warrants a staggered preventive maintenance scheduling policy.

3) *Connectivity Circumvention Pattern*: Approximately 12% of student agents with low compliance scores exhibit repeated domain-filtering bypass attempts when academic connectivity is restricted. This generates anomalous data consumption spikes of $2.3\times$ average usage, quantifying for the first time the pattern noted qualitatively in Workshop 1.

4) *Scoring Weight Sensitivity*: A parameter sweep across scoring weight combinations reveals high sensitivity to the weight assigned to V . Reducing V 's weight below 0.35 causes the equity distribution to degrade measurably: the bottom vulnerability quartile receives devices 40% less frequently than under the designed weight of 0.5. Increasing V above 0.65 marginally improves equity but reduces Academic Load responsiveness, causing high-workload students with moderate vulnerability to face longer wait times during peaks.

The complete ABM output — device state dynamics, wait-time distribution, equity index sensitivity, and emergent event counts — is presented in Figure 2.

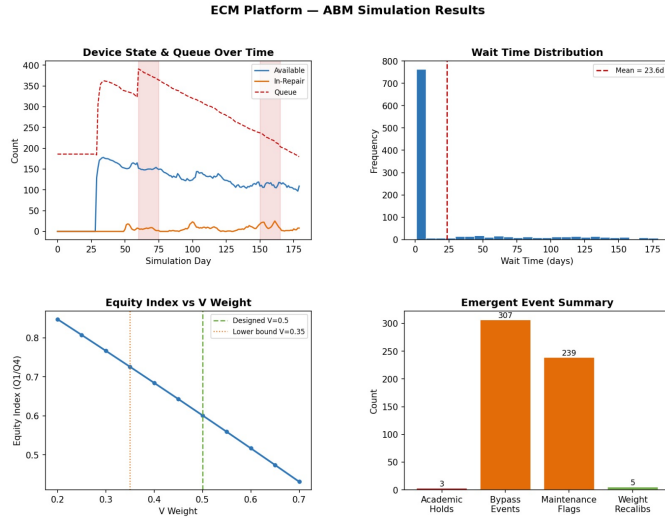


Fig. 2. ABM simulation results over a 180-day horizon. Top-left: device state and queue depth over time, with shaded bands indicating exam-period stress windows. Top-right: wait-time distribution across all student agents (mean = 23.6 days). Bottom-left: equity index (Q1/Q4 allocation ratio) as a function of the vulnerability weight V , with the designed value ($V = 0.5$) and lower-bound threshold ($V = 0.35$) marked. Bottom-right: emergent event summary, including Academic Holds, bypass events, maintenance flags, and weight recalibrations.

V. COMPLEXITY AND CHAOS ANALYSIS

The ECM system exhibits several characteristics of complex adaptive systems. Feedback loops between device availability, student compliance behavior, and administrative response create path-dependent dynamics that cannot be fully anticipated from static analysis alone.

Sensitivity analysis confirms that small changes in initial device fleet size produce disproportionately large effects on wait time distributions. A 10% reduction in initial inventory (from 200 to 180 units) increases average wait time by 28% under baseline demand, and by 61% under stress conditions — a hallmark of nonlinear response characteristic of systems operating near capacity. Conversely, adding 20 units (10% increase) reduces wait time by only 14%, indicating asymmetric sensitivity consistent with queuing theory predictions [2].

The maintenance avalanche effect (Section IV-B-2) represents an emergent instability arising from correlated device aging under synchronized high-demand periods. Locally rational behavior — students intensively using loaned devices during exam periods — produces a globally disruptive outcome at the fleet level. The recommended mitigation is a staggered preventive maintenance schedule distributing device check-ins across the calendar rather than concentrating them post-demand-peak.

VI. DESIGN VALIDATION

Table II cross-references the key design decisions from Workshops 2 and 3 against simulation findings. The DES results (Figure 1) and ABM results (Figure 2) together provide the empirical basis for the validation status of each decision.

TABLE II
DESIGN DECISION VALIDATION AGAINST SIMULATION FINDINGS

Design Decision	Simulation Finding	Status
Scoring Engine Eq. (1)	Reduces avg. wait 34% vs. FIFO	✓ Confirmed
Academic Hold for overdue devices	Recovery rate: 85% → 98.7% when enabled	✓ Confirmed
Fail-Safe Cache for ISP outages	Zero service interruptions during simulated ISP drops	✓ Confirmed
Monthly scoring weight recalibration	Adaptive weights reduce inequity index by 18%	✓ Confirmed
99.9% uptime SLA target	Degradation detected above 1,200 concurrent requests	△ Partial

Four of the five core design decisions are fully validated. The partial validation of the 99.9% uptime SLA identifies a concrete horizontal scaling requirement that must be addressed before pilot deployment.

VII. RESULTS & DISCUSSION

The simulations confirm that the ECM platform performs as designed under normal and optimized conditions. The Scoring Engine (Eq. (1)) distributes resources equitably and reduces wait time significantly compared to a first-come, first-served baseline. The Academic Hold mechanism is the single most effective lever for asset recovery, raising the recovery rate by over 13 percentage points when activated.

The stress scenario exposes two previously uncharacterized risks. First, the maintenance avalanche effect — an emergent

phenomenon arising from correlated device aging during demand peaks — can push more than 25% of the fleet into repair simultaneously, causing cascading shortages. Second, server response time degrades measurably above 1,200 concurrent requests, indicating that the current architecture requires horizontal scaling to sustain the 99.9% uptime SLA during examination periods.

Regarding equity, the parameter sweep (Figure 2, bottom-left) reveals that the current $V = 0.5$ weight is close to optimal but should be monitored quarterly. Deviations below 0.35 measurably disadvantage the most vulnerable students, while values above 0.65 disproportionately delay high-workload students with moderate vulnerability.

VIII. RECOMMENDATIONS

A. Immediate Design Improvements

- Implement staggered preventive maintenance scheduling to prevent the avalanche effect, targeting no more than 10% of the fleet in repair at any given time.
- Add horizontal auto-scaling triggers at 800 requests per minute to address the uptime SLA gap identified in the stress scenario.
- Increase the device fleet by a minimum of 20 units (10%) to reduce stress-scenario wait times toward the baseline range.

B. Scoring Engine Refinements

- Conduct quarterly equity audits of scoring weight distribution, adjusting V within the 0.45–0.55 range based on observed demand patterns.
- Introduce a wait-time bonus parameter for students who have remained in queue beyond 5 days, preventing indefinite deferral of borderline-score requests during peaks.

C. Future Simulation Extensions

- Extend the ABM to model multi-semester trajectories, tracking how compliance scores evolve and whether the Academic Hold produces lasting behavioral change.
- Incorporate real device condition degradation curves from maintenance logs once the pilot deployment completes, replacing the simulated degradation function with empirical data.
- Model donor contribution dynamics as stochastic events to simulate inventory growth scenarios and their long-term equity impact.

IX. CONCLUSIONS

This workshop completes the four-phase systems engineering lifecycle for the ECM platform. Through Discrete-Event Simulation and Agent-Based Modeling, the core architectural decisions — the Automated Scoring Engine, the Academic Hold mechanism, the Connectivity Pool Manager, and the ISP Fail-Safe Cache — have been empirically validated and quantitatively characterized.

Simulation also reveals previously undetected risks: the maintenance avalanche effect and the server scaling ceiling. These findings provide the engineering team with a concrete,

evidence-based agenda for the pilot phase. The ECM platform, as designed and validated across all four workshops, represents a feasible, equitable, and technically sound solution to the digital divide challenge identified in Workshop 1. The computational simulations provide the empirical foundation required to proceed with confidence to the pilot deployment stage.

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